

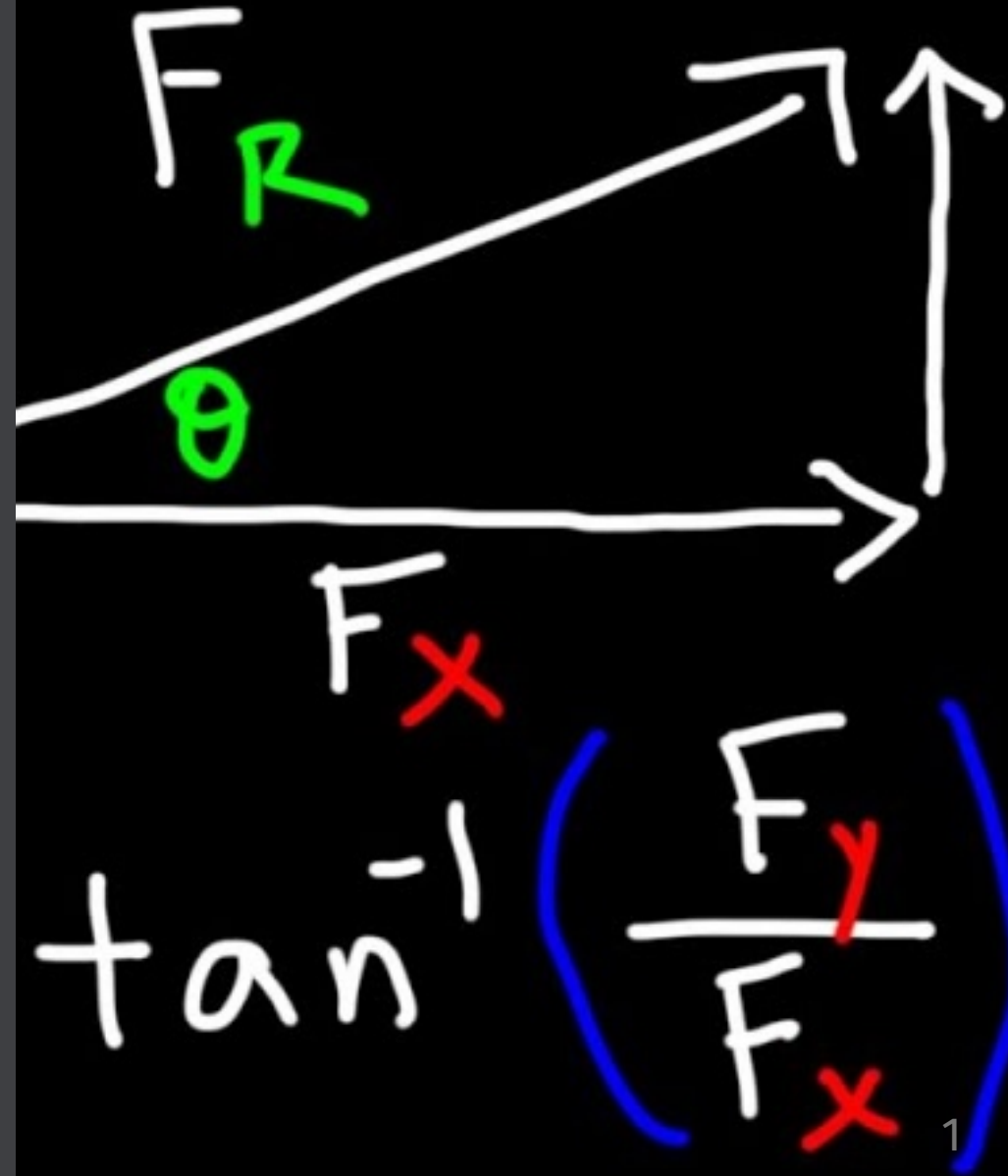
Day 04 - Models & Modeling

$$\mathbf{r} = r_x \hat{x} + r_y \hat{y} + r_z \hat{z}$$

$$\mathbf{r} = r_x \hat{i} + r_y \hat{j} + r_z \hat{k}$$

$$\mathbf{r} = r_x \hat{e}_1 + r_y \hat{e}_2 + r_z \hat{e}_3$$

$$\mathbf{r} = \langle r_x, r_y, r_z \rangle$$



Announcements

- Homework 1 is due next Friday
- Help sessions will start next week
- Last class was not recorded properly
- No class on Monday (MLK Day)
- Complete the [student information poll](#) by Monday, please

Goals for this week

Be able to answer the following questions.

- What is Classical Mechanics?
- How can we formulate it?
- What are the essential physics models for single particles?
- What mathematics do we need to get started?

Reminders from Day 02

- Classical Physics has existed for centuries
- Newton's Laws are one formulation of Classical Mechanics
- We can use Newton's Laws to describe the motion of particles
- Newton's Laws give rise to differential equations

